

PIYUSH KUMAR SINGH

Student

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Summary

Motivated Computer Science student specializing in Data Science at Manipal University Jaipur, with strong foundations in Python programming, data structures, SQL, and data analysis. Experienced in developing data-driven applications, including an AI-based financial advisory chatbot that leverages machine learning and natural language processing techniques. Interested in applying data science, machine learning, and analytical skills to solve complex real-world problems and build intelligent systems.

Skills

Programming & Querying: Python, C Programming, Data Structures and Algorithms, SQL (Basic)

Data Analysis Libraries: Pandas, NumPy

Experience

Arduino Workshop

Dec 2022 – Dec 2022

Microcontroller Programming

Tools Used: Arduino IDE, Arduino Uno, Sensors & Electronic Components

- **Designed and developed a prototype automated fish feeding machine using Arduino to support animal welfare by ensuring timely feeding.**
- Programmed the Arduino microcontroller using Arduino IDE (C/C++) to control the feeding mechanism and automate feeding intervals.
- Built and tested electronic circuits using a breadboard, sensors, and basic components for reliable system operation.
- Integrated hardware and software components to create a functional prototype demonstrating automated feeding for aquaculture environments.

Projects

Financial Advisory Chatbot | Python, NLP

January 2025 – Present

- Developing an AI-based financial advisory chat bot to answer user queries and provide investment guidance using machine learning techniques.
- Implementing XGBoost models to analyze financial data and predict risk scores or suitable investment categories.
- Using XGBRanker to rank investment options such as stocks, bonds, and mutual funds based on user profiles and market data.
- Applying SHAP and LIME techniques to explain model predictions and improve transparency in financial recommendations.
- Designing a system architecture that integrates natural language processing, predictive modeling, ranking algorithms, and explainable AI.

Awards & Certifications

- **Google AI Essentials Certificate – Coursera (2025):** Learned fundamentals of artificial intelligence, machine learning concepts, and practical AI applications.
- **Programming in Java Certificate - NPTEL (2025):** Recognized for strong Java programming skills and problem-solving ability.
- **Dean's List – Manipal University Jaipur (2 Times):** Awarded for outstanding academic performance and maintaining a high CGPA.

Education

B.Tech Computer Science (Data Science)

Graduated: 2028

Manipal University Jaipur, Jaipur

CGPA: 8.7 / 10